

# Constructing and Solving Equations — Paper A

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every equation on this paper has a whole-number solution unless it says otherwise.
- Whatever you do to one side, **do to the other**. Write the operation beside each line, like “ $-4$ ” or “ $\div 3$ ”.
- Strip the numbers off the  $x$  first and **divide last**. Dividing early makes you divide every term.
- Every answer can be checked by putting it back into the **original** equation. Do it every time — it costs ten seconds and it catches nearly every error.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – ONE STEP, THEN TWO

- 1.** Complete the table. Each empty box is worth one mark. In the last column, show the substitution that proves your answer is right. [6]

| Equation      | Solution | Check (substitute back) |
|---------------|----------|-------------------------|
| $x + 9 = 17$  |          |                         |
| $5x = 45$     |          |                         |
| $3x + 2 = 20$ |          |                         |

The check column is not decoration — it is where you find out whether you were right before the marker does.

- 2.** Solve:

**(a)**  $x - 6 = 11$  [1]

**(b)**  $7x = 56$  [1]

**(c)**  $x \div 4 = 6$  [1]

**(d)**  $x + 13 = 5$

[1]

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**3.** Solve:

**(a)**  $2x + 5 = 17$

[1]

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**(b)**  $4x - 3 = 25$

[1]

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**(c)**  $5x + 8 = 8$

[1]

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**(d)**  $3x - 11 = 4$

[1]

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SECTION 2 – BOTH SIDES, AND BRACKETS

**4.** Solve, showing each step:

**(a)**  $5x = 2x + 18$

[2]

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**(b)**  $7x - 4 = 3x + 12$

[2]

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**5.** Solve:

**(a)**  $2(x + 5) = 16$

[2]

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**(b)**  $3(x - 4) = 21$

[2]

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SECTION 3 – BUILDING THE EQUATION YOURSELF

**6.** I think of a number, multiply it by 5, then subtract 3. The result is 32.

**(a)** Using  $n$  for the number, form an equation.

[1]

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**(b)** Solve it.

[1]

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**(c)** Check your answer against the original instructions.

[1]

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**7.** A rectangle has width  $x$  cm and length  $(x + 3)$  cm. Its perimeter is 26 cm.

**(a)** Form an equation in  $x$ .

[1]

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**(b)** Solve it.

[1]

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**(c)** Write down the width and the length, and check the perimeter.

[1]

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**8.** Ali has  $n$  marbles. Ben has 7 more than Ali. Together they have 41 marbles.

**(a)** Form an equation in  $n$ .

[1]

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**(b)** Solve it, and state how many marbles each boy has.

[1]

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*End of paper. Check every answer has a unit or a form the question actually asked for.*